

Unit 8

REACTION KINETICS

38 items · 127 marks

Syllabus sections in this unit

8.1 Rate of reaction

8.2 Effect of temperature on reaction rates and the concept of activation energy

8.3 Homogeneous and heterogeneous catalysts

Unit 8: Reaction kinetics**1. M/J 26/P21/Q2/b(ii)**

2 The Contact process uses sulfur to produce sulfuric acid.

stage 1 Sulfur is burned in excess air.

stage 2 $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g}) \quad \Delta H = -196 \text{ kJ mol}^{-1}$

stage 3 SO_3 reacts to produce H_2SO_4 .

(b) The reaction conditions for stage 2 include:

- using a ratio of 1 mol SO_2 to 1 mol O_2
- heat in the presence of V_2O_5 .

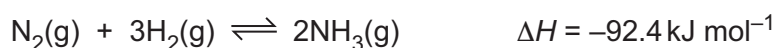
(ii) Explain how V_2O_5 affects the rate of reaction.

[2]

2. M/J 26/P22/Q1/e(ii)

1 Oxygen reacts with many elements.

(e) Ammonia is made in the Haber process.



A temperature of 400–450 °C is used in the Haber process.

(ii) Describe how the initial rate of the forward reaction changes, if at all, when a temperature greater than 450 °C is used. Explain your answer.

Assume all other variables remain constant.

change in initial rate

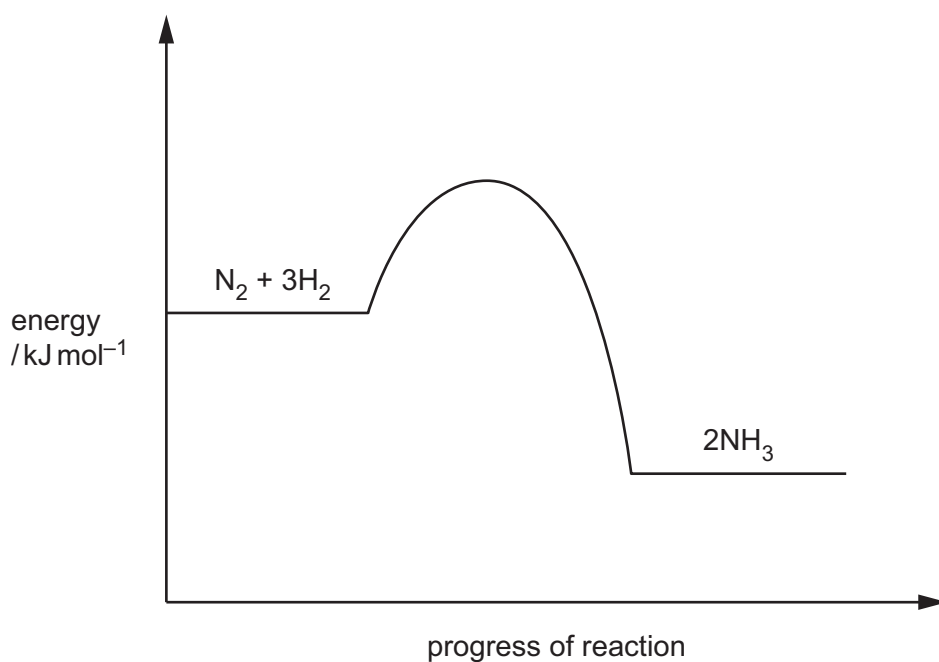
explanation

[3]

3. M/J 26/P24/Q2/b(i),d,e

(b) Nitrogen is used to manufacture ammonia, NH_3 , in the Haber process. An iron catalyst is used.

(i) Complete the reaction pathway diagram in Figure 2.1 to show the effect of adding an iron catalyst to the reaction of nitrogen and hydrogen.

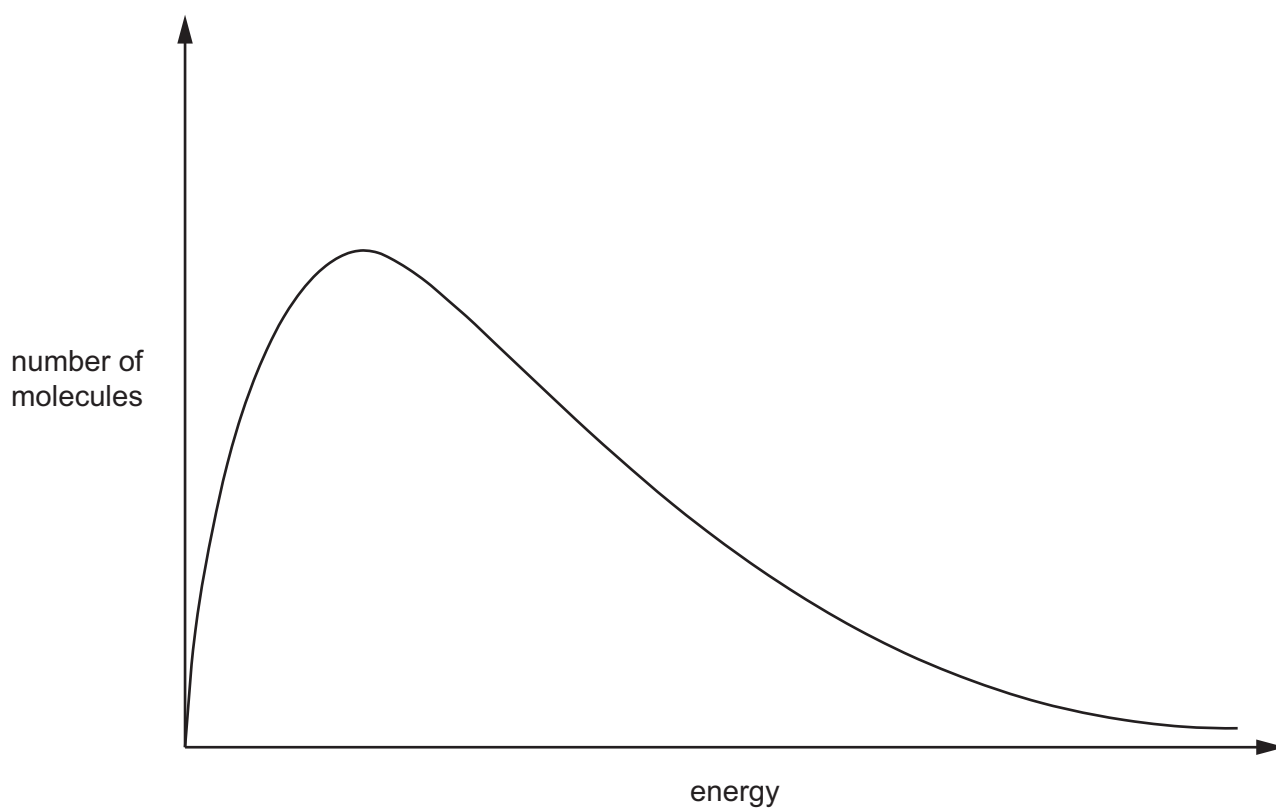
**Figure 2.1**

(d) Explain the effect of increasing temperature on rate of reaction.

[1]

(e) Figure 2.2 shows the Boltzmann distribution for a sample of X(g) at temperature T_1 .

[3]

**Figure 2.2**

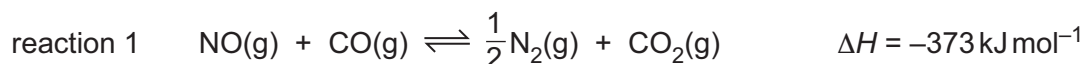
Sketch a curve on Figure 2.2 to show the Boltzmann distribution for the sample of X(g) at a temperature higher than T_1 .

[2]

4. MAR 26/P22/Q2/d(ii)

2 Nitrogen monoxide, NO, is an atmospheric pollutant.

(d) Reaction 1 occurs during the catalytic removal of NO from car exhaust gases.



The activation energy, E_a , of this reaction is 570 kJ mol^{-1} .

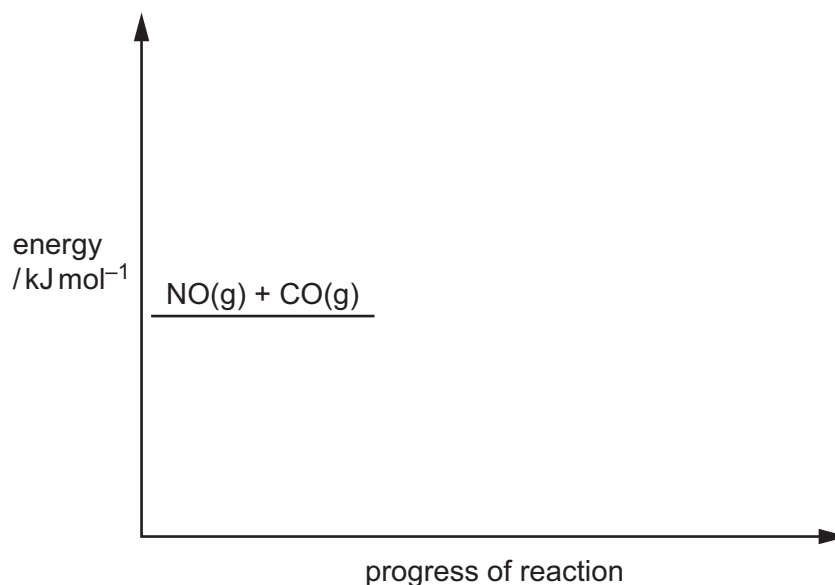


Figure 2.3

(ii) On your diagram in Figure 2.3, show how the addition of a catalyst affects reaction 1. [1]

5. O/N 25/P24/Q2/c(i)

2 Aluminium oxide, Al_2O_3 , and phosphorus(V) oxide, P_4O_{10} , are both used as reagents and catalysts.

(b) A 5.00 dm^3 sealed flask contains 0.400 mol of CO(g) and 0.800 mol of $\text{H}_2\text{(g)}$ and an Al_2O_3 catalyst. The flask is heated to a temperature of 290°C and allowed to reach equilibrium. Equation 1 shows the reaction.



The equilibrium constant, K_c , of equation 1 is given.

$$K_c = \frac{[\text{CH}_3\text{OH}]}{[\text{CO}][\text{H}_2]^2}$$

(c) P_4O_{10} catalyses the reversible reaction of CO with H_2 to form CH_3OH .



P_4O_{10} then acts as a dehydrating agent, causing CH_3OH to form CH_3OCH_3 .

(i) Explain how the presence of a catalyst affects a chemical reaction.

[1]

6. M/J 25/P21/Q3/a

- 3 (a)** Hexene reacts with hydrogen gas to produce hexane.

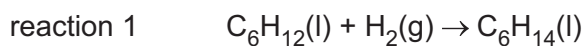


Fig. 3.1 shows the distribution of energies of $\text{H}_2(\text{g})$ molecules at temperature T .

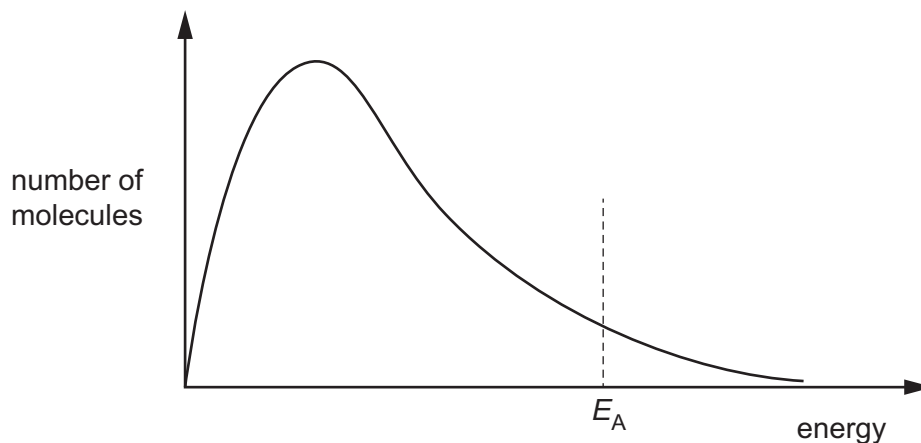


Fig. 3.1

- (i) Sketch on Fig. 3.1 the shape of the curve for the same sample of $\text{H}_2(\text{g})$ molecules when the temperature is increased. [2]
- (ii) Explain why increasing the temperature increases the rate of reaction 1. [2]
- (iii) State the role of nickel when it is added to reaction 1. [1]
- (iv) Annotate Fig. 3.2 to show the effect of adding nickel to reaction 1 at temperature T .

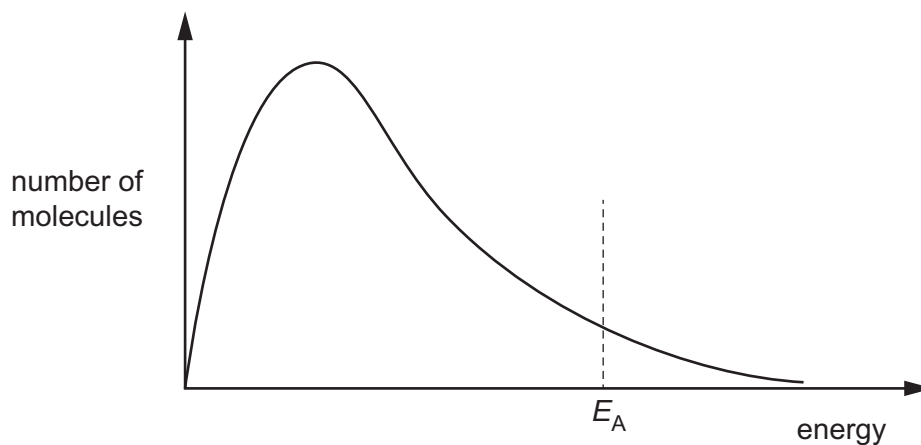
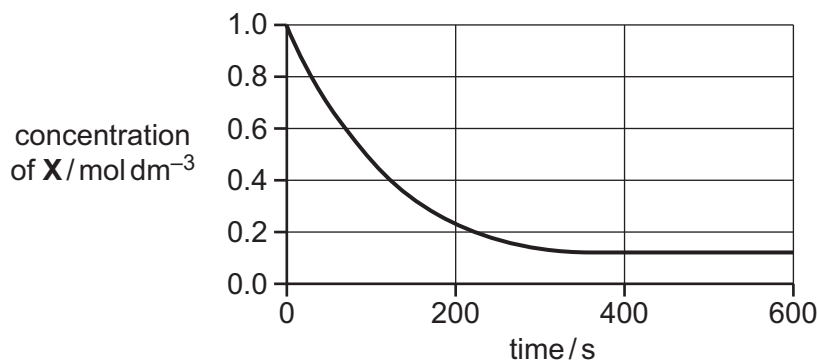


Fig. 3.2

[1]

7. M/J 25/P22/Q3/b,d

- (b) $\text{H}_2(\text{g})$ and nickel are added to alkene **X**. Fig. 3.2 shows how the concentration of **X** changes with time.

**Fig. 3.2**

- (i) Use Fig. 3.2 to identify the limiting reagent in this reaction. Explain your answer.

[1]

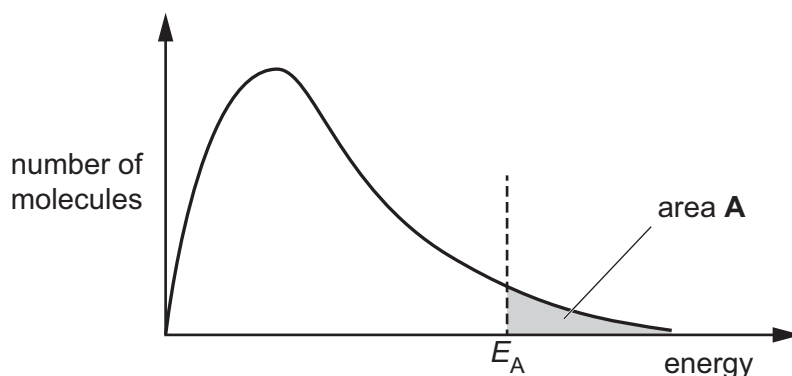
- (ii) Use Fig. 3.2 to describe how the gradient changes as the reaction proceeds. State what this shows about the rate during this reaction.

[1]

- (d) Hydrocarbon **Y** reacts with $\text{H}_2(\text{g})$.

Fig. 3.3 shows the distribution of energies of $\text{H}_2(\text{g})$ at temperature T .

Area **A** represents the number of molecules with energy greater than or equal to the activation energy, E_A , at temperature T .

**Fig. 3.3**

- (i) Annotate Fig. 3.3 to show the effect of adding nickel to **Y** and $\text{H}_2(\text{g})$ at temperature T . [1]
- (ii) Area **B** (not labelled on Fig. 3.3) represents the number of molecules with energy greater than or equal to the activation energy when nickel is added at temperature T .

State the difference, if any, between areas **A** and **B**. Explain the significance of your answer on the rate of hydrogenation of **Y**. Give your answer in terms of collisions.

[2]

8. M/J 25/P23/Q3/b(ii,iii)

(b) C_4H_8 reacts with an excess of H_2 to produce C_4H_{10} .



(ii) Define activation energy, E_{A} .

[1]

(iii) The Boltzmann distribution for the reaction mixture in reaction 1 is shown in Fig. 3.1. Use the Boltzmann distribution to explain the effect of adding a catalyst on the rate of reaction.

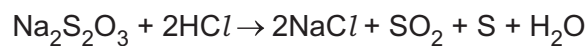


Fig. 3.1

[2]

9. M/J 25/P23/Q4/a

4 The reaction between $\text{Na}_2\text{S}_2\text{O}_3(\text{aq})$ and $\text{HCl}(\text{aq})$ is monitored at constant temperature.



10. M/J 24/P23/Q2/b(ii)

2 $\text{N}_2(\text{g})$ reacts with $\text{H}_2(\text{g})$ in the Haber process, as shown in reaction 1.

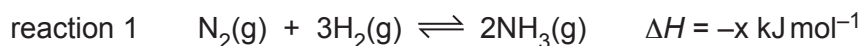


Table 2.1 shows the different conditions used to produce three equilibrium mixtures, **A**, **B** and **C**.

Table 2.1

	A	B	C
initial molar ratio of $\text{N}_2 : \text{H}_2$ added	1 : 3	1 : 3	1 : 3
temperature / °C	500	500	1000
pressure / atm	1000	1000	1000
iron present in mixture	no	yes	no
percentage yield of $\text{NH}_3(\text{g})$ at equilibrium	58	x	y

- (ii) Describe and explain the change to the rate of the forward reaction that occurs to establish the equilibrium in **C** compared to **A**.

You do **not** need to refer to the Boltzmann distribution in your answer.

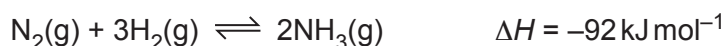
[2]

11. MAR 24/P22/Q3/c(ii,iii)

3 Nitrogen, N_2 , is generally an unreactive molecule but it does react under certain conditions.

- (c) The Haber process involves the reaction of N_2 and H_2 to form ammonia, NH_3 .

A catalyst is used, which allows the process to be carried out at a lower temperature and pressure.



- (ii) Explain how the presence of a catalyst affects the reaction.

[1]

- (iii) State and explain the effect, if any, on the **rate** of the Haber process as the pressure is lowered.

[2]

12. O/N 23/P21/Q3/b

3 Potassium chlorate, KClO_3 , is widely used as an oxidising agent and to make $\text{O}_2(\text{g})$.

(b) $\text{KClO}_3(\text{s})$ decomposes when heated.

$\text{MnO}_2(\text{s})$ catalyses the exothermic decomposition reaction.

Complete and label the diagram in Fig. 3.1 to show the effect of $\text{MnO}_2(\text{s})$ on the decomposition of $\text{KClO}_3(\text{s})$.

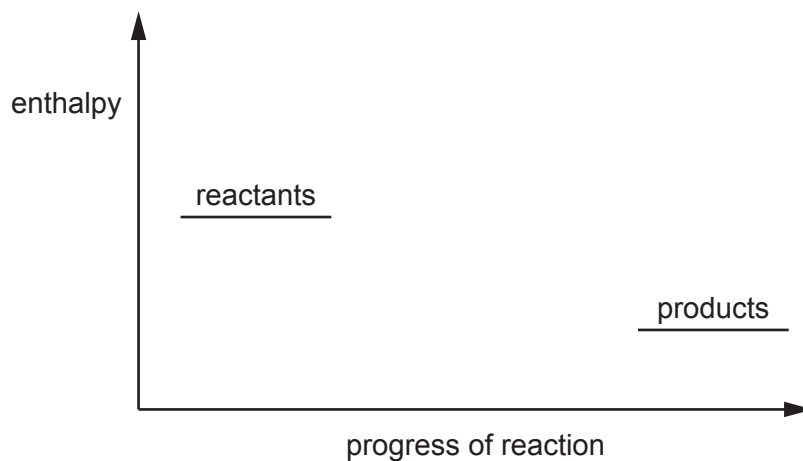


Fig. 3.1

[2]

13. O/N 23/P23/Q3/b

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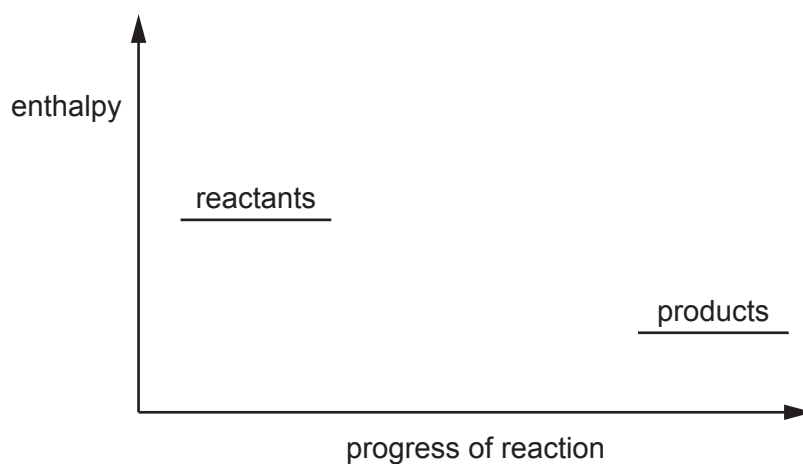
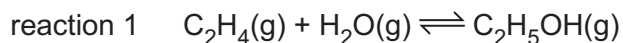


Fig. 3.1

[2]

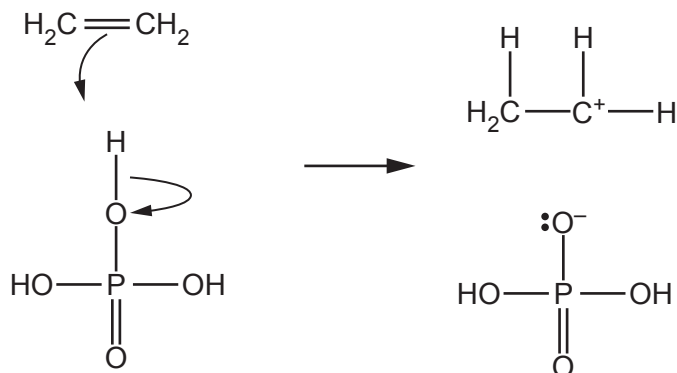
14. M/J 23/P22/Q4/c(iv,v)

4 In industry, ethanol is made by reacting ethene with steam in the presence of H_3PO_4 .



(c) The mechanism for reaction 1 can be described in three steps. Steps 1 and 2 for reaction 1 are shown in Fig. 4.1.

step 1



step 2

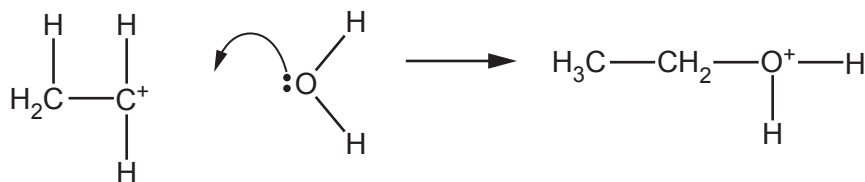


Fig. 4.1

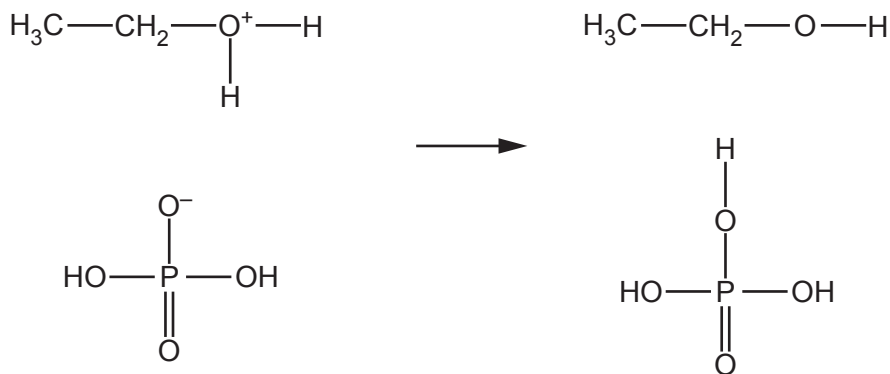


Fig. 4.2

(iv) Describe how a catalyst affects a reaction. Explain your answer.

[2]

(v) Use Fig. 4.1 and Fig. 4.2 to justify why H_3PO_4 is described as a catalyst in reaction 1.

[1]

15. MAR 23/P22/Q3/d(i)

- 3** Alkenes undergo an addition reaction with a 1:1 mixture of CO and H₂ to form aldehydes.

Fig. 3.1 shows the reaction of propene with a 1:1 mixture of CO and H₂.

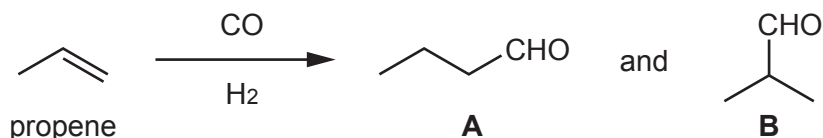
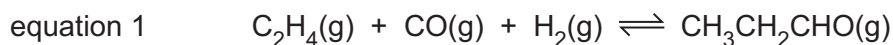


Fig. 3.1

- (d)** The reaction of ethene, C₂H₄, with a 1:1 mixture of CO and H₂ is shown in equation 1.



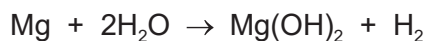
At atmospheric pressure a cobalt-based catalyst is used in this reaction.

- (i)** State and explain the effect of using a catalyst on this reaction.

[2]

16. M/J 22/P21/Q1/g

- 1** Calcium, magnesium and radium are Group 2 elements. Radium follows the same trends as the other members of Group 2.
- (g)** Cold water reacts slowly with a piece of Mg to produce bubbles of H₂(g).
Cold water reacts rapidly with burning Mg to produce H₂(g) in an explosive mixture.



Explain why the rate of reaction of cold water with burning magnesium is greater.

[2]

17. M/J 22/P23/Q3/a(iv,v)

3 **G** belongs to a group of compounds called ethers.

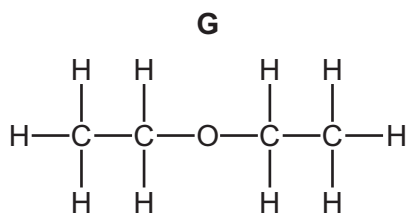
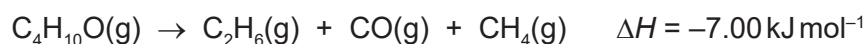


Fig. 3.1

(a) When **G**, $C_4H_{10}O$, is heated, thermal decomposition occurs.



(iv) Thermal decomposition of **G** in the presence of I_2 affects the activation energy, E_a , for the reaction. Table 3.2 shows E_a for the thermal decomposition of **G** with and without I_2 .

Table 3.2

reaction	E_a (with I_2)/kJ mol ⁻¹	E_a /kJ mol ⁻¹
$C_4H_{10}O(g) \rightarrow C_2H_6(g) + CO(g) + CH_4(g)$	143	224

State what effect adding I_2 to the reaction mixture has on the value of K_c .
Explain your answer.

[2]

(v) Fig. 3.2 shows the Boltzmann distribution of energies for molecules of **G** at constant temperature, $T^\circ\text{C}$.

Sketch, on Fig. 3.2, the Boltzmann distribution of energies for molecules of **G** at a higher temperature, $(T+100)^\circ\text{C}$.

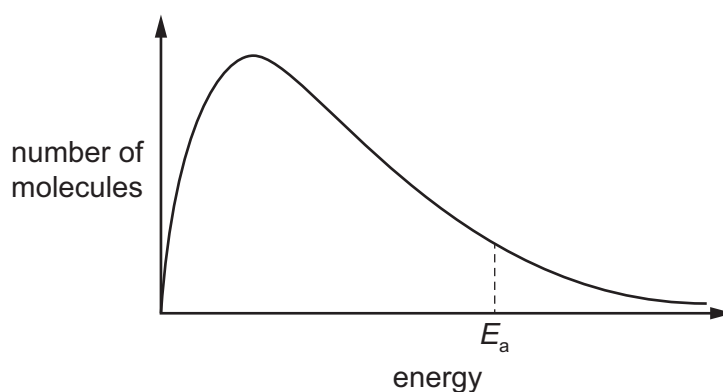


Fig. 3.2

[2]

18. MAR 22/P22/Q2/b(iii)

2 Some oxides of elements in Period 3 are shown.



(b) Al_2O_3 is an amphoteric oxide found in bauxite.

(iii) Al_2O_3 is used as a catalyst in the dehydration of alcohols.

State the effect of using Al_2O_3 as a catalyst in the dehydration of alcohols. Use the Boltzmann distribution in Fig. 2.1 to help explain your answer.

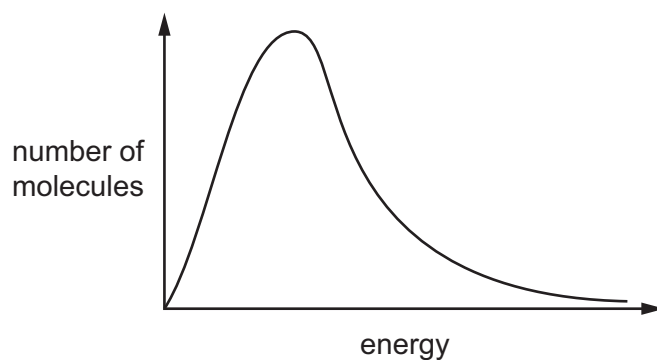
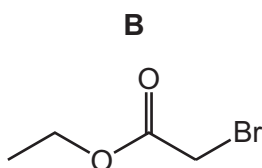


Fig. 2.1

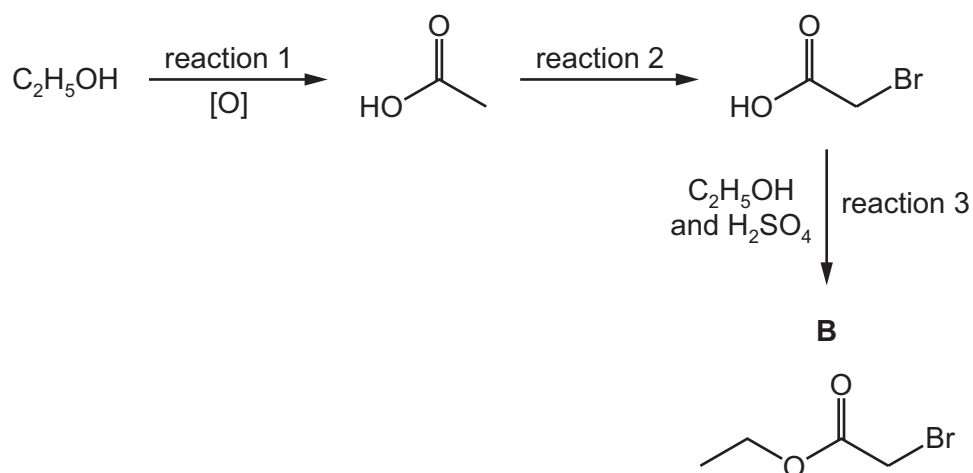
[3]

19. O/N 21/P21/Q4/a(iv)

4 Compound **B** is a liquid with a fruity smell.



The reaction scheme shows how **B** can be made from ethanol, $\text{C}_2\text{H}_5\text{OH}$.



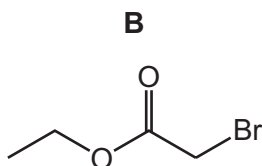
(iv) H_2SO_4 acts as a homogeneous catalyst in reaction 3.

Explain why H_2SO_4 is described as *homogeneous*.

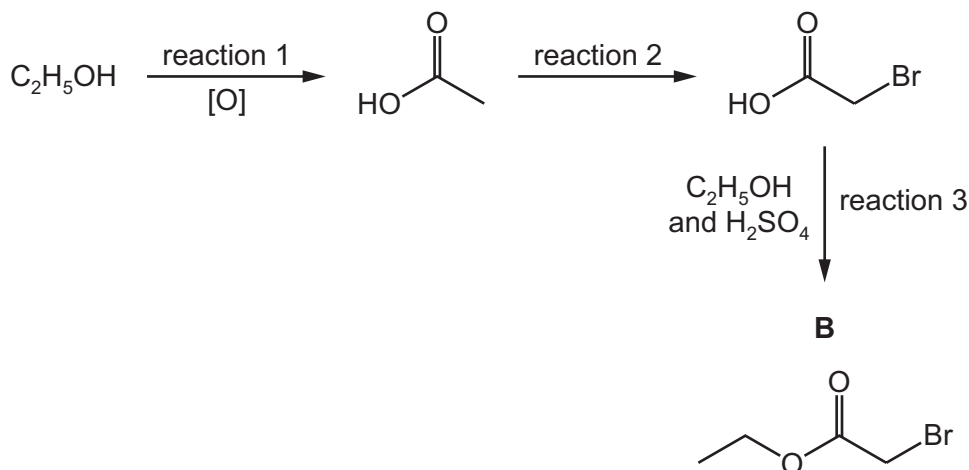
[1]

20. O/N 21/P23/Q4/a(iv)

- 4 Compound **B** is a liquid with a fruity smell.



The reaction scheme shows how **B** can be made from ethanol, $\text{C}_2\text{H}_5\text{OH}$.



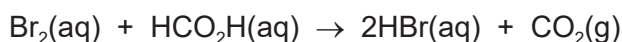
- (iv) H_2SO_4 acts as a homogeneous catalyst in reaction 3.

Explain why H_2SO_4 is described as *homogeneous*.

[1]

21. M/J 21/P21/Q4/b,c

- 4 Aqueous bromine reacts with methanoic acid to form hydrogen bromide and carbon dioxide gas.



The table shows the oxidation numbers of bromine and carbon in the species involved in this reaction.

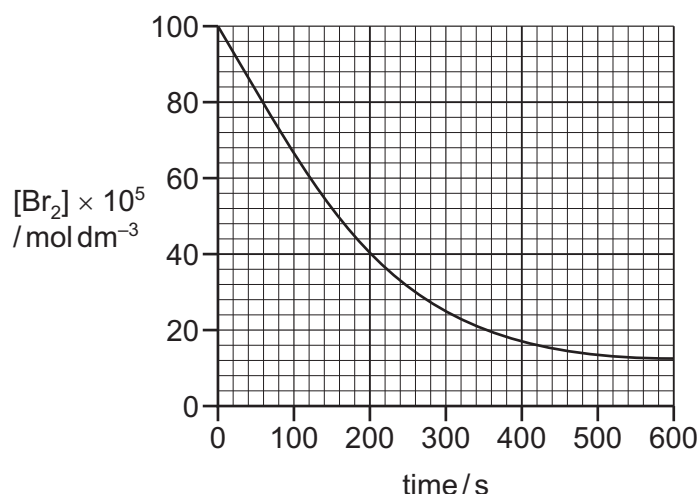
	Br in Br_2	C in HCO_2H	Br in HBr	C in CO_2
oxidation number	0	+2	-1	+4

- (b) Suggest one change you would observe, ignoring temperature changes, when bromine reacts with methanoic acid.

[1]

- (c) This reaction can be followed by measuring the concentration of bromine present in the mixture at regular time intervals.

The graph shows the change in concentration of bromine against time in a reaction carried out at 20°C .



- (i) Use the graph to calculate the average rate of reaction at 20 °C during the first 600 s. State the units of this rate of reaction.

average rate of reaction

units

[2]

The experiment is repeated at a temperature of 40 °C. This relatively small increase in temperature produces a large increase in reaction rate.

- (ii) Sketch a graph, on the same axes, to show the expected results when repeating the experiment at 40 °C. [1]
- (iii) The rate of reaction increases when the frequency of successful collisions between reactant particles increases.

Explain why an increase in temperature produces this effect.

[2]

22. M/J 21/P22/Q3/a,b

- 3** A large excess of 2-bromo-2-methylpropane is added to 0.0010 mol of NaOH(aq), which contains a few drops of phenolphthalein indicator. A stopwatch is started as soon as the substances are mixed. The time taken for the pink colour to disappear is recorded.

The experiment is repeated at different temperatures, keeping all concentrations and volumes of reagents constant.

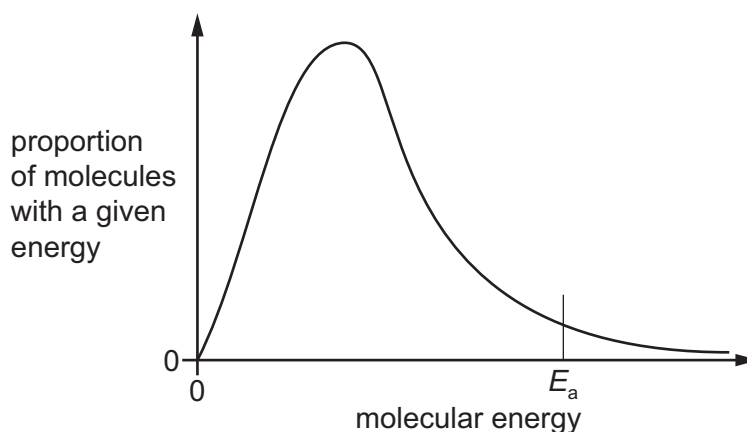
temperature / °C	time taken for pink colour to disappear / s
20	300
25	65
35	20

- (a)** Explain what is meant by the term *rate of reaction*.

[1]

- (b)** The graph shows the energy distribution of molecules in a sample of 2-bromo-2-methylpropane at 25 °C.

E_a represents the activation energy for the reaction.



- (i)** Label the graph to show the proportion of 2-bromo-2-methylpropane molecules which have sufficient energy to react. [1]
- (ii)** Use the same axes to sketch the distribution of energies of molecules in a sample of 2-bromo-2-methylpropane at 50 °C. [2]
- (iii)** State the effect of an increase in temperature on E_a for this reaction.

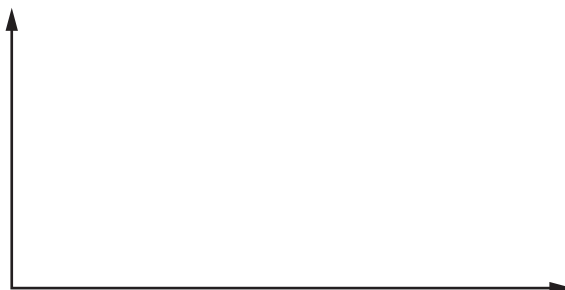
[1]

23. MAR 21/P22/Q1/a,c(ii)

1 The rate of chemical reactions is affected by changes in temperature and pressure.

- (a) (i)** Draw a curve on the axes to show the Boltzmann distribution of energy of particles in a sample of gaseous krypton atoms at a given temperature.

Label the curve **T1** and label the axes.



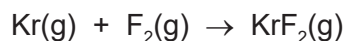
[2]

- (ii)** On the diagram in **(a)(i)**, draw a second curve to show the distribution of energies of the krypton atoms at a higher temperature.

Label the second curve **T2**.

[1]

- (c)** Krypton reacts with fluorine in the presence of ultraviolet light to make krypton difluoride, $\text{KrF}_2(\text{g})$.



activation energy for the reaction, $E_a = +385 \text{ kJ mol}^{-1}$

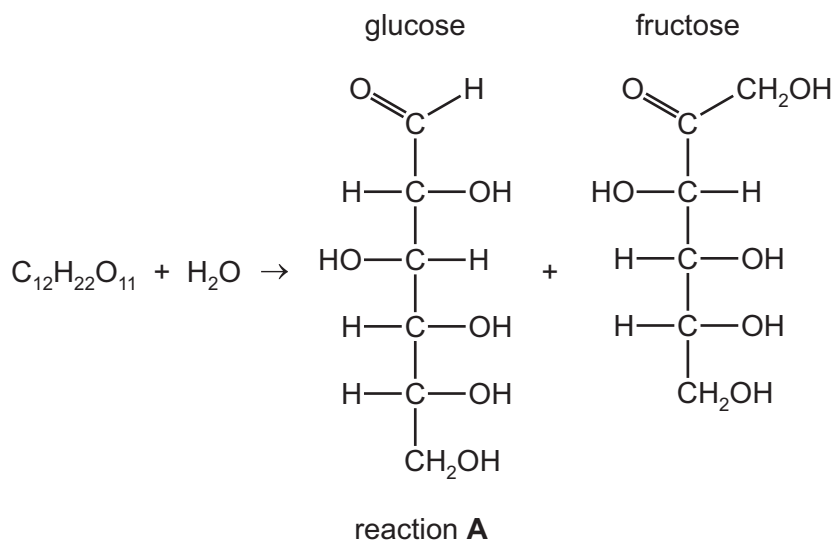
enthalpy change of formation of KrF_2 , $\Delta H_f = +60.2 \text{ kJ mol}^{-1}$

- (ii)** Explain, in terms of activation energy, E_a , and the collision of particles, how an increase in temperature affects the rate of a chemical reaction.

[2]

24. M/J 20/P22/Q3/c

3 Sucrose, $C_{12}H_{22}O_{11}$, reacts with water to form glucose and fructose in reaction **A**.



(c) Reaction **A** occurs faster in the presence of an enzyme. This is reaction **B**.

(i) The activation energy for reaction **B** is $+29 \text{ kJ mol}^{-1}$.

Predict a value for the activation energy of reaction **A**.

[1]

(ii) The enthalpy change for reaction **A** is -14 kJ mol^{-1} .

Predict a value for the enthalpy change for reaction **B**.

[1]

(iii) Sketch a labelled energy level diagram for reaction **B**. Use relevant values from **(c)(i)** and **(c)(ii)**.



[2]

25. MAR 20/P22/Q1/c

1 Group 2 metals form alkaline solutions in water.

- (c) Oxygen reacts readily with some metals, but each Group 2 metal requires strong heating to start the reaction with oxygen.

Suggest why strong heating is required to start these reactions.

[1]

26. MAR 20/P22/Q2/e(i)

2 The Group 17 elements, chlorine, bromine and iodine, are non-metals that show trends in their physical and chemical properties.

- (e) Lucas's reagent is a mixture of HCl and ZnCl_2 . Primary, secondary and tertiary alcohols can be distinguished by their reaction with Lucas's reagent.

Alcohols react with the HCl in Lucas's reagent to form halogenoalkanes.

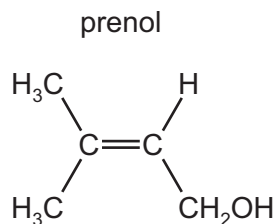
ZnCl_2 acts as a homogeneous catalyst for these reactions.

- (i) Explain the meaning of the term *homogeneous*.

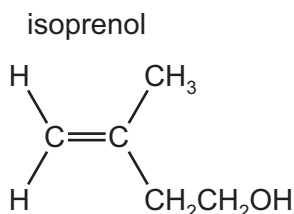
[1]

27. O/N 19/P22/Q4/d(ii)

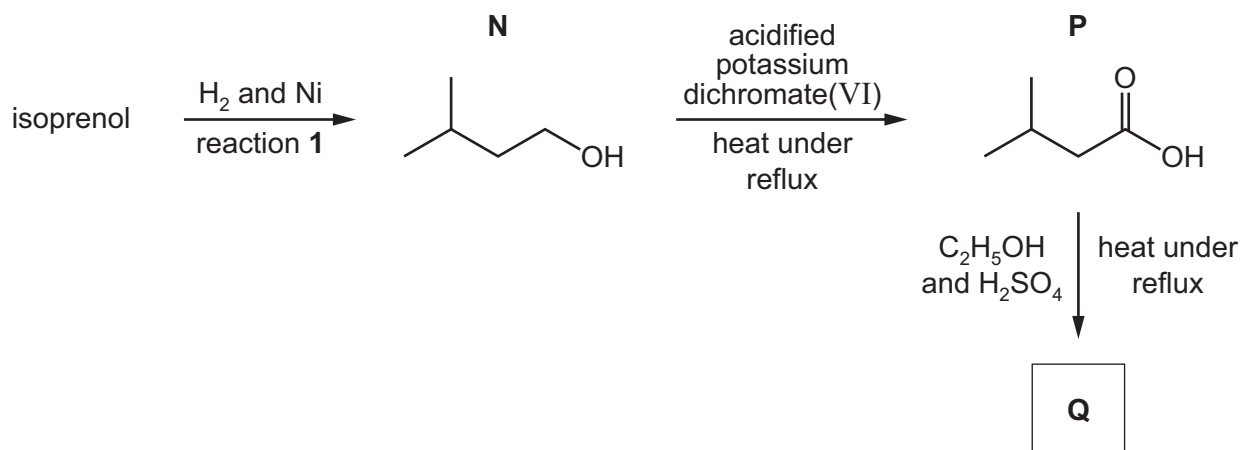
- 4 Prenol is a naturally occurring organic molecule found in many fruits. It contains both an alkene and an alcohol functional group.



- (d) Isoprenol is a structural isomer of prenol.



The series of reactions shows how isoprenol can be used to form **Q**, a sweet-smelling liquid.



- (ii) Isoprenol is a liquid.

Ni acts as a catalyst for reaction 1.

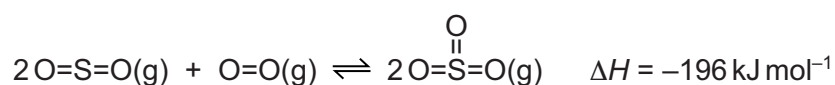
Identify the type of catalysis shown by Ni in reaction 1.

[1]

28. M/J 18/P21/Q1/a,c,d(i)

- 1 Sulfuric acid is manufactured by the Contact process.

One stage in this process is the conversion of sulfur dioxide into sulfur trioxide in the presence of a heterogeneous catalyst of vanadium(V) oxide, V_2O_5 .



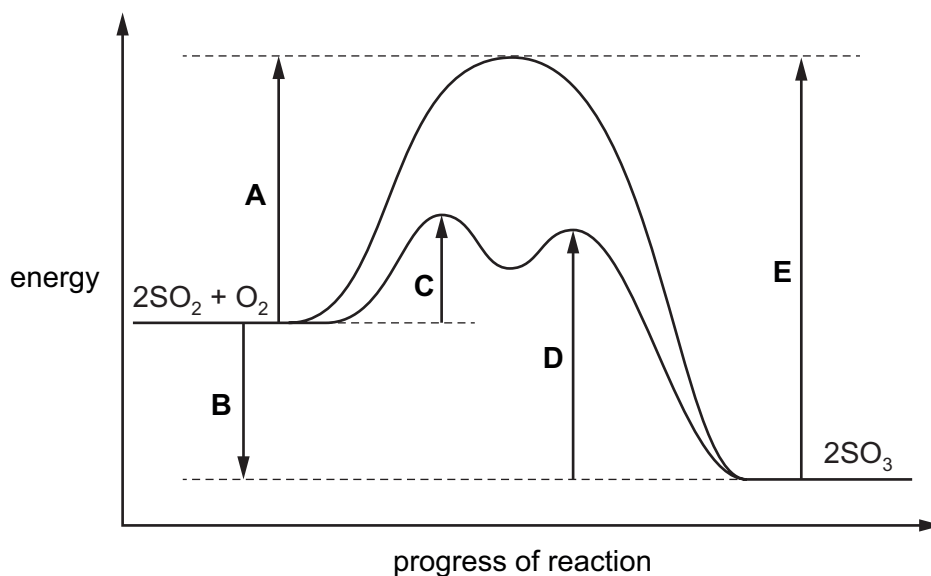
- (a) (i) State the effect of a catalyst on a reaction.
Explain how a catalyst causes this effect.

[2]

(ii) State the meaning of the term *heterogeneous* as applied to catalysts.

[1]

(c) A reaction pathway diagram for both the catalysed and uncatalysed reactions between SO_2 and O_2 is shown.



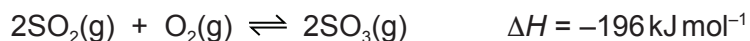
The letters **A–E** represent energy changes.

Complete the table by stating which letter, **A–E**, represents the energy change described.

energy change	letter
the energy change for the production of SO_3	
the activation energy for the production of SO_3 in the absence of a catalyst	
the activation energy for the first step in the decomposition of SO_3 in the presence of a catalyst	

[3]

The equation for this stage of the Contact Process is shown.



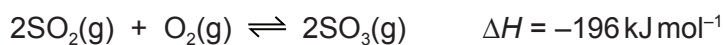
(d) (i) State and explain the effect of increasing temperature on the rate of production of SO_3 .

[3]

29. M/J 18/P23/Q1/c(ii,iv)

1 The elements sodium to chlorine, in the third period, all form oxides.

(c) SO_3 is produced by the reaction between SO_2 and O_2 in the Contact process. A dynamic equilibrium is established.



(ii) Explain why the gradients of the SO_2 and O_2 lines decrease with time.

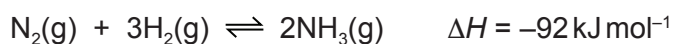
[2]

(iv) Suggest a reason why the initial gradient of the SO_2 line is steeper than that of the O_2 line.

[1]

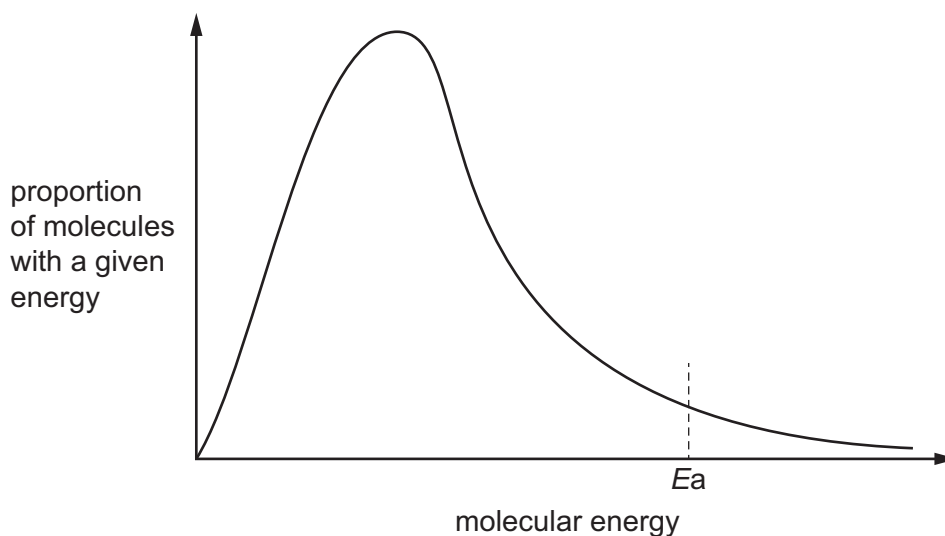
30. O/N 17/P21/Q1/b(i,ii)

1 Ammonia, NH_3 , is manufactured from nitrogen and hydrogen by the Haber process.



(b) The Haber process is usually carried out at a temperature of approximately 400°C in the presence of a catalyst. Changing the temperature affects both the rate of production of ammonia and the yield of ammonia.

The Boltzmann distribution for a mixture of nitrogen and hydrogen at 400°C is shown. E_a represents the activation energy for the reaction.



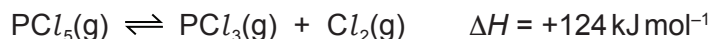
(i) Using the same axes, sketch a second curve to indicate the Boltzmann distribution at a higher temperature. [2]

(ii) **With reference to the Boltzmann distribution**, state and explain the effect of increasing temperature on the rate of production of ammonia.

[3]

31. O/N 17/P22/Q2/b(i)

- 2 At 450 K phosphorus(V) chloride, $\text{PCl}_5(\text{g})$, decomposes to form phosphorus(III) chloride, $\text{PCl}_3(\text{g})$, and chlorine, $\text{Cl}_2(\text{g})$. A dynamic equilibrium is established as shown.

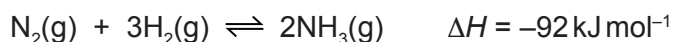


- (b) (i) State and explain the effect of increasing temperature on the rate of decomposition of $\text{PCl}_5(\text{g})$.

[2]

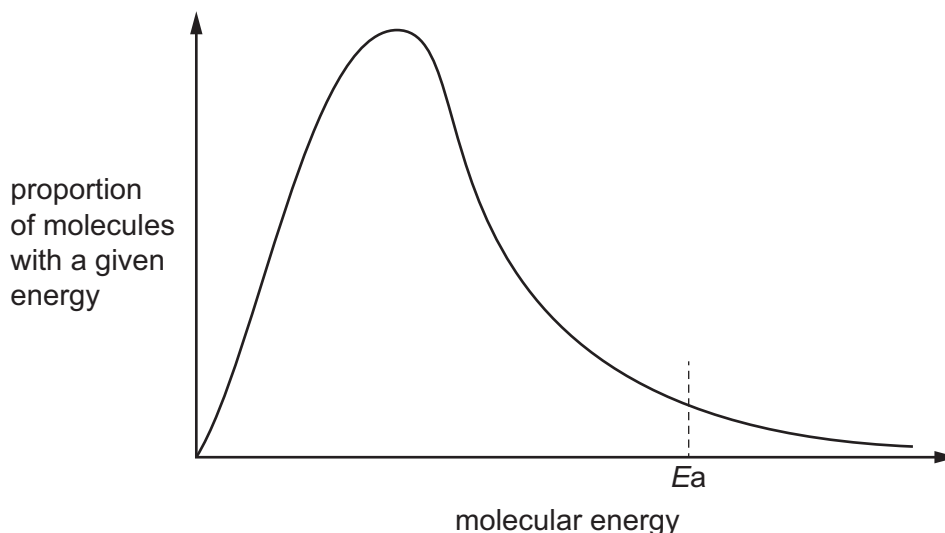
32. O/N 17/P23/Q1/b(i,ii)

- 1 Ammonia, NH_3 , is manufactured from nitrogen and hydrogen by the Haber process.



- (b) The Haber process is usually carried out at a temperature of approximately 400°C in the presence of a catalyst. Changing the temperature affects both the rate of production of ammonia and the yield of ammonia.

The Boltzmann distribution for a mixture of nitrogen and hydrogen at 400°C is shown. E_a represents the activation energy for the reaction.

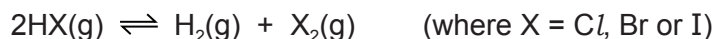


- (i) Using the same axes, sketch a second curve to indicate the Boltzmann distribution at a higher temperature. [2]
- (ii) **With reference to the Boltzmann distribution**, state and explain the effect of increasing temperature on the rate of production of ammonia.

[3]

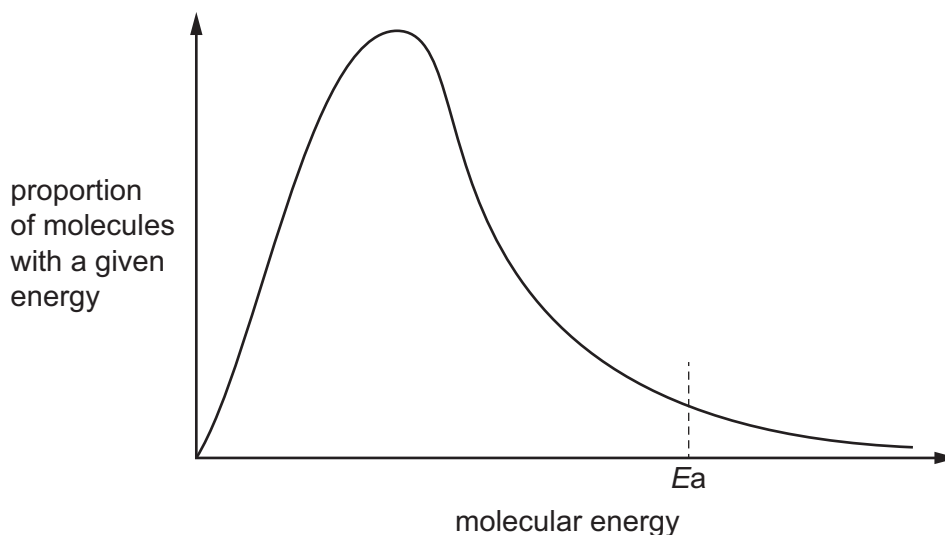
33. M/J 17/P22/Q3/b(i,ii)

- 3** The hydrogen halides, HCl, HBr and HI, can undergo thermal decomposition. In a sealed container an equilibrium is established according to the equation shown.



- (b)** At a temperature of 700 K a sample of HBr is approximately 10% decomposed. Changing the temperature affects both the rate of decomposition of HBr and the percentage that decomposes.

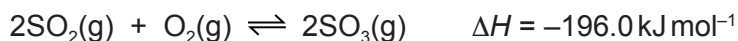
The Boltzmann distribution for a sample of HBr at 700 K is shown. E_a represents the activation energy for the reaction.



- (i)** Using the same axes, sketch a second curve to indicate the Boltzmann distribution at a higher temperature. [2]
- (ii)** With reference to the curves, state and explain the effect of increasing temperature on the **rate** of decomposition of HBr. [3]

34. M/J 17/P23/Q3/b(i,ii)

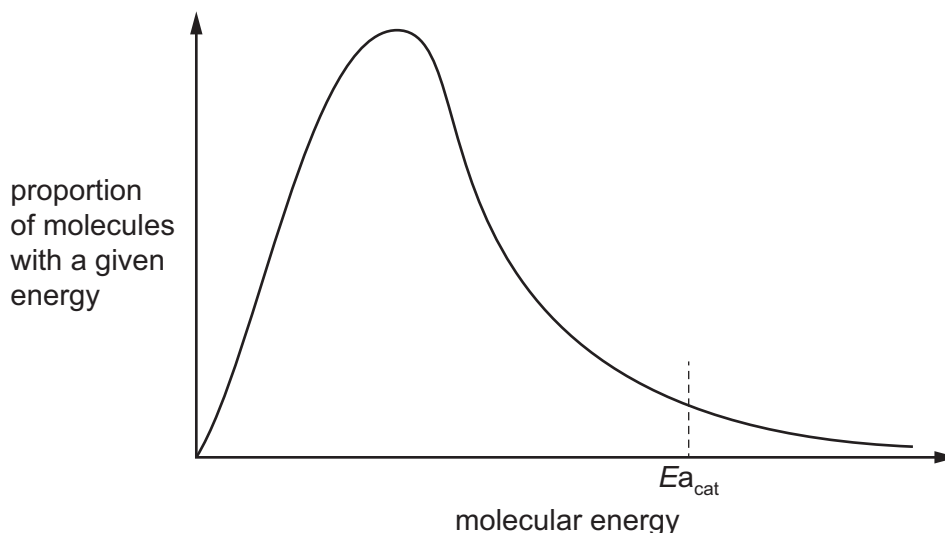
- 3** Sulfur trioxide, SO_3 , is manufactured from sulfur dioxide and oxygen by the Contact process.



- (b)** The Contact process is usually carried out at a temperature of approximately 700 K, a pressure of approximately 150 kPa and in the presence of a vanadium(V) oxide catalyst, V_2O_5 .

The Boltzmann distribution for a mixture of SO_2 and O_2 at 700 K is shown.

$E_{\text{a cat}}$ represents the activation energy for the reaction in the presence of the catalyst.



- (i)** Add a labelled mark, $E_{\text{a uncat}}$, to the diagram to indicate the activation energy in the absence of the catalyst. [1]
- (ii)** State the benefit of using a catalyst in this reaction. Explain how it achieves this effect. [2]

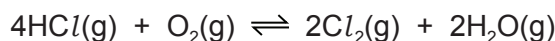
35. MAR 17/P22/Q2/c(ii)

- 2** Hydrogen halides are compounds formed when halogens (Group 17 elements) react with hydrogen. The bond polarity of the hydrogen halides decreases from HF to HI.

Some relevant data are shown in the table.

hydrogen halide	HF	HCl	HBr	HI
boiling point/ $^{\circ}\text{C}$	19	-85	-67	-35
H-X bond energy/ kJ mol^{-1}	562	431	366	299

- (c)** Hydrogen chloride undergoes a reversible reaction with oxygen.



The reaction is carried out at 400°C in the presence of a copper(II) chloride catalyst.

- (ii)** State the **type** of catalyst used in this reaction. Explain how a catalyst is able to increase the rate of a chemical reaction. [2]

36. M/J 16/P21/Q3/b(ii)

3 The elements in Group 2, and their compounds, show many similarities and trends in their properties.

(b) Magnesium oxide can be formed by the reaction of magnesium and oxygen in the air.

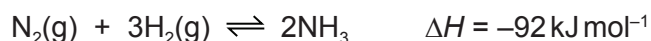
(ii) Explain why there is no visible reaction when a piece of magnesium ribbon is exposed to the air.

[2]

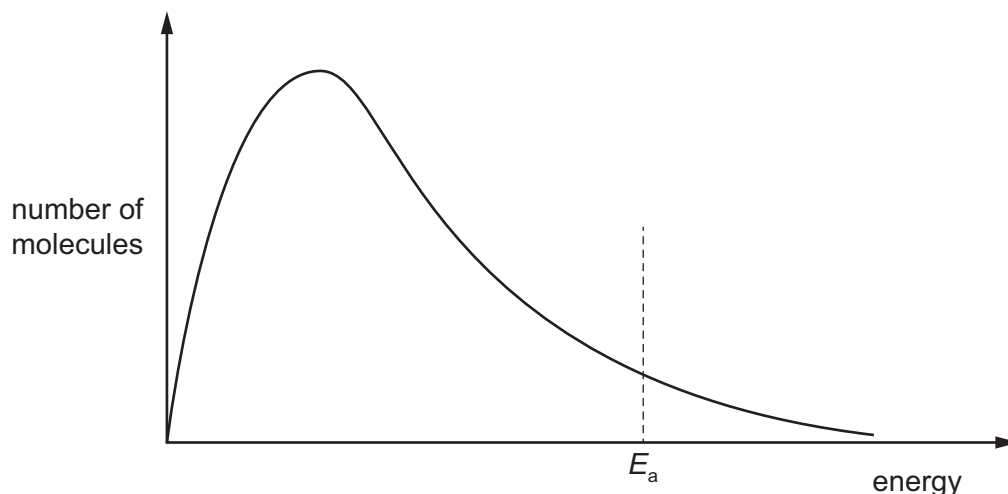
37. M/J 16/P23/Q2/b(i,ii,iii)

2 Ammonium nitrate is an important fertiliser made by the acid-base reaction between ammonia and nitric acid.

(b) The Haber process involves a reaction between nitrogen and hydrogen at a temperature of 450 °C and a pressure of 20 000 kPa. At a higher reaction temperature, the rate of production of ammonia would be greater.



The Boltzmann distribution curve shows the distribution of energies in a mixture of nitrogen and hydrogen at 450 °C.



(i) Sketch a second line onto the axes above to show the distribution of energies in the same mixture of gases at a higher temperature. [2]

(ii) With reference to the two curves, explain why the rate of production of ammonia would be greater at a higher temperature.

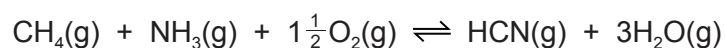
[2]

(iii) Add a suitable label to the horizontal axis above and use it to explain why a catalyst is used in the Haber process.

[2]

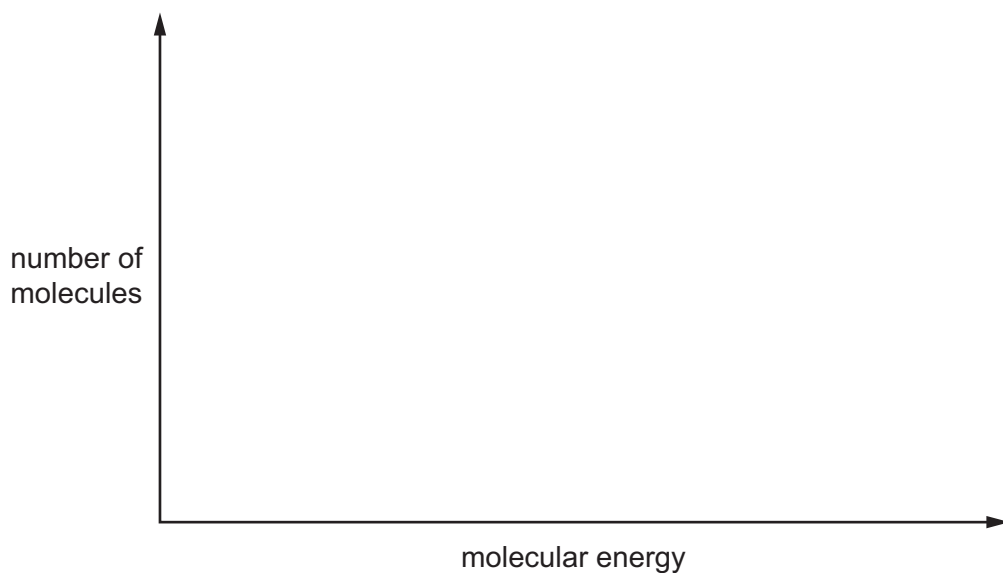
38. MAR 16/P22/Q3/c

- 3** Over one million tonnes of hydrogen cyanide, HCN, are produced each year using the Andrussow process. The overall equation for the reaction is shown.



- (c)** The process uses a platinum catalyst, which increases the rate of reaction.

Sketch a Boltzmann distribution on the axes given below and use your diagram to explain how the platinum catalyst increases the rate of the reaction.



[3]

Answers Section

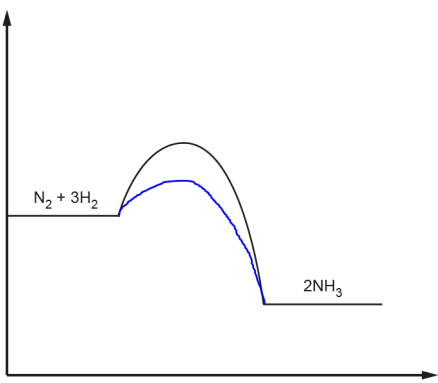
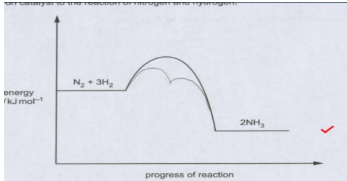
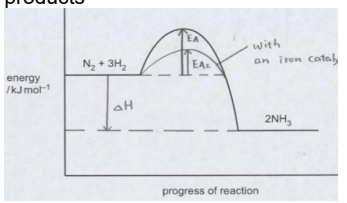
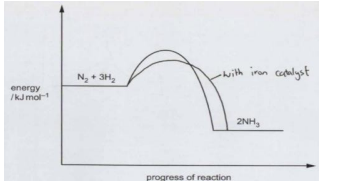
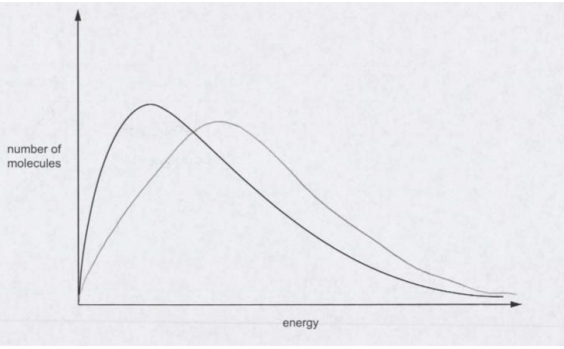
1. M/J 26/P21/Q2/b(ii)

2(b)(ii)	M1 it / rate (of reaction) increases M2 (it) lowers EA OR (used) to lower EA	2 rate = speed successful = effective particles = molecules M1 A greater frequency of successful collisions / more successful collisions per unit time / higher successful collision rate / more frequent successful collisions / more successful collisions per second A for per = / M2 A more O₂ and/or SO₂ (molecules) OR more particles have sufficient / enough / minimum energy to react M2 A more O₂ and / or SO₂ (molecules) OR more particles have energy $\geq$ EA A Ea / EA / AE for EA I more of the collisions have energy greater than or equal to activation energy I pathway (even if wrong) I redox I it is a catalyst
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2. M/J 26/P22/Q1/e(ii)

1(e)(ii)	change in rate explanation M1 increases M2 more particles have $E \geq E_A$ M3 so greater frequency of effective collisions.	3 I yield / Le Chatelier arguments M1 might be seen in explanation if not on first answer line M1 A it takes less time to reach equilibrium M2 A more particles have energy greater than / equal to the activation energy / E_A M2 R particles have more activation energy M2 I particles have more energy M2 I atoms / reactants M3 A more effective collisions per unit time / increased rate of effective collisions M3 I higher chance / probability of more successful collisions M3 I more collisions / more effective collisions alone effective = successful particles = molecules activation energy = sufficient / enough energy to react
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3. M/J 26/P24/Q2/b(i),d,e

2(b)(i)		1. A BOD double hump for cat complex intermediate  A line that joins original line before products  A BOD 
2(d)	M1 (increasing temperature) increases rate M2 increased number of particles/collisions with E_A OR larger proportion of particles/collisions have energy greater than or equal to E_A / $E \geq E_A$ M3 increased / higher frequency of effective / successful collisions	3 M1 A BOD increases speed of reaction M2 A more particles with enough / sufficient energy to react M3 A increased rate of effective / successful collisions A increase in effective / successful collisions per unit time A 'increased frequency of collisions' AND 'more effective / successful collisions' I increased chance / probability of effective / successful collision I more collisions (only) I references to Le Chatelier / yield A particles = molecules = atoms
2(e)	 • start at origin and not crossing original until beyond the peak • peak to the RHS of original peak AND lower than original • line does not cross the original after the peak reached AND crosses the original once only ✓✓	2 A BOD slight upwards 'kick' at the end. Max 1 completely correct graphs for temperature $< T_1$

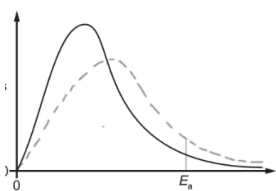
4. MAR 26/P22/Q2/d(ii)

2(d)(ii)	diagram has lower-energy transition state(s)	1
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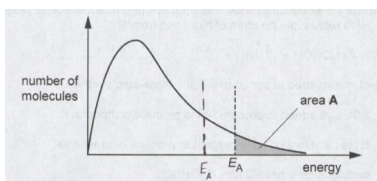
5. O/N 25/P24/Q2/c(i)

2(c)(i)	increases rate by providing a different mechanism with a lower activation energy	1
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6. M/J 25/P21/Q3/a

3(a)(i)	 Two correct for one mark, all four for two marks: • start at origin • peak sits below the original • peak sits to the right-hand side of the original • new line crosses the original once only AND to the right of the original peak	2
3(a)(ii)	M1 greater proportion of (H_2) molecules have energy greater than or equal to the activation energy $\geq E_A$ M2 increase in frequency of successful collisions (between H_2 and C_6H_{12})	2
3(a)(iii)	catalyst	1
3(a)(iv)	annotation to show new E_A (nickel / catalyst) further to the left than original E_A	1

7. M/J 25/P22/Q3/b,d

3(b)(i)	H_2 / hydrogen AND some X / alkene left(over) / remains (at the end of this reaction)	1
3(b)(ii)	<i>link the significance of gradient values to rate AND describe either the decrease in rate or the decrease in gradient</i> as the gradient decreases the rate decreases OR rate decreases until the reaction stops when gradient = 0	1
3(d)(i)	E_A shown on Fig. 3.3 to the left of original (on the x axis) 	1
3(d)(ii)	M1 it / (area) B larger AND rate (reaction / hydrogenation) increases OR (area) B identified by shading or labelling on Fig 3.3 AND rate (reaction / hydrogenation) increases in prose M2 increase in the frequency of effective / successful collisions OR increase in rate of effective / successful collisions	2

8. M/J 25/P23/Q3/b(ii,iii)

3(b)(ii)	minimum energy required for a collision to be effective	1
3(b)(iii)	M1 • area under the curve with energy $\geq E_A$ using a catalyst is larger so • greater proportion of particles with sufficient energy to react M2 rate faster because higher frequency of successful collisions	2

9. M/J 25/P23/Q4/a

4(a)(i)	Check final graph for range of acceptable answers • $\pm(1-0.5)/100 = \pm 0.005$ • 0 • $\text{mol dm}^{-3} \text{s}^{-1}$ for both Any 2 points = 1 mark, all 3 points = 2 marks	2
4(a)(ii)	$\text{Na}_2\text{S}_2\text{O}_3$ because reaction stops before all the HCl is used up. OR $\text{Na}_2\text{S}_2\text{O}_3$ because there is still HCl remaining (when the reaction has stopped)	1
4(a)(iii)	the frequency of effective collisions decreases as concentration of HCl(aq) and / or $\text{Na}_2\text{S}_2\text{O}_3$ decreases	1
4(a)(iv)	M1 0.001(00) mol sulfur produced M2 $M1 \times 6.02 \times 10^{23}$ $= 6.02 \times 10^{20}$	2
4(a)(v)	It reacts with water (to produce H_2SO_3)	1

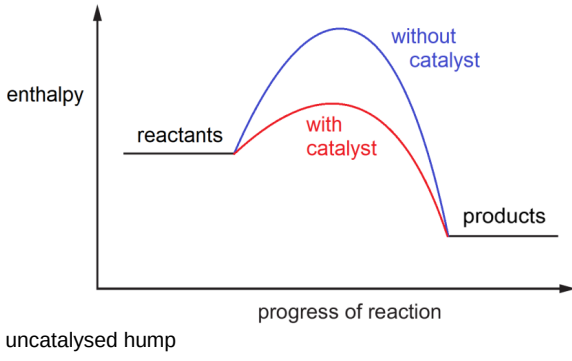
10. M/J 24/P23/Q2/b(ii)

2(b)(ii)	(increase in temperature / energy from A to C (so)) M1 more collisions with particles that have $E > E_a$ AND increase frequency of effective collisions M2 (initial) rate (of the forward reaction) increases	2
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11. MAR 24/P22/Q3/c(ii,iii)

3(c)(ii)	increases rate by providing a different mechanism with a lower activation energy	1
3(c)(iii)	rate is lowered	1
	lower frequency of successful collisions / number of successful collisions per unit time	1

12. O/N 23/P21/Q3/b

3(b)	 enthalpy reactants without catalyst with catalyst products progress of reaction uncatalysed hump catalysed hump lower than uncatalysed	1
		1

13. O/N 23/P23/Q3/b

3(b)		1
	enthalpy reactants without catalyst with catalyst products progress of reaction uncatalysed hump catalysed hump lower than uncatalysed	1

14. M/J 23/P22/Q4/c(iv,v)

4(c)(iv)	M1 a catalyst increases rate of reaction M2 by producing an alternative reaction pathway with lower activation energy	2
4(c)(v)	it is regenerated OR it is reformed	1

15. MAR 23/P22/Q3/d(i)

3(d)(i)	increases the rate of reaction	1
	by providing an alternative reaction pathway of lower E_a	1

16. M/J 22/P21/Q1/g

1(g)	M1 (heat / energy released from burning Mg) provides more particles with energy $\geq E_a$ M2 frequency of successful / effective collisions is greater	2
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17. M/J 22/P23/Q3/a(iv,v)

3(a)(iv)	M1 no change / none M2 I_2 is a catalyst AND does not affect position of equilibrium OR I_2 is a catalyst AND reduces time taken to reach (same) equilibrium	2
3(a)(v)	- start at origin and not crossing original until beyond the peak - peak to the right-hand side of original peak AND lower than original - line does not cross the original after the peak reached	2

18. MAR 22/P22/Q2/b(iii)

2(b)(iii)	two lines shown on diagram, e.g. E_A and $E_{A,cat}$ [1] greater proportion of molecules with $E \geq E_A$ [1] frequency of effective collisions increases [1]	3
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19. O/N 21/P21/Q4/a(iv)

4(a)(iv)	in the same phase / in same state	1
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20. O/N 21/P23/Q4/a(iv)

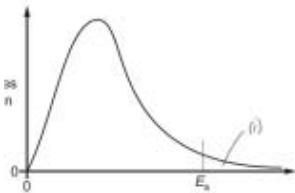
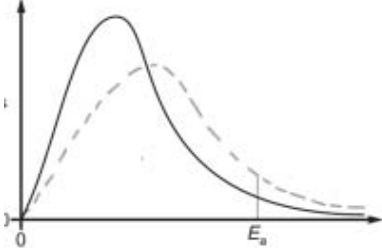
4(a)(iv)	in the same phase / in same state	1
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21. M/J 21/P21/Q4/b,c

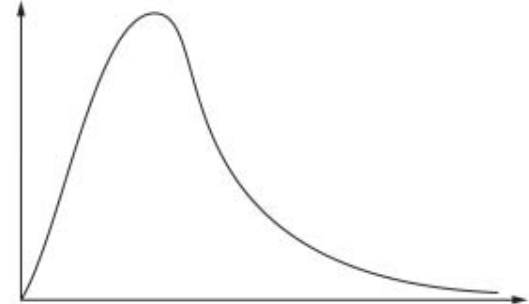
4(b)	(solution) turns (from brown / orange / red to) colourless / decolorises OR brown / orange / red fades	1
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4(c)(i)	rate = total change in concentration of Br ₂ divided by time taken <i>calculation dependent on graph</i> $(100 \times 10^{-5} - 12 \times 10^{-5}) / 600$ M1 average rate of reaction 1.47×10^{-6}	1
	M2 units mol dm ⁻³ s ⁻¹	1
4(c)(ii)	graph shown on same axes has steeper initial gradient AND reaches the same final [Br ₂]	1
4(c)(iii)	M1 (at increased temp the average kinetic) energy of particles / species / molecules increases.	1
	M2 (many) more/greater proportion of particles with energy $\geq E_a$	1

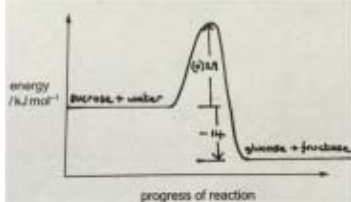
22. M/J 21/P22/Q3/a,b

3(a)	change in amount of substance with time	1
3(b)(i)	label area under the original curve to the right of E _a . 	1
3(b)(ii)	M1 curve starts at the origin but peak lies to the right of original.	1
	M2 peak at higher temperature is lower than the original AND graph crosses the original once only – beyond the peak of original 	1
3(b)(iii)	no change / none	1

23. MAR 21/P22/Q1/a,c(ii)

1(a)(i)	vertical axis: number of particles  horizontal axis: (kinetic / particle) energy M1 : shape of curve correct M2 : labelled axes	2
1(a)(ii)	Labelled line (T2) with lower peak to right of original	1
1(c)(ii)	- rate increases (increase in temperature means) more particles have energy $\geq$ activation energy - frequency of successful collisions increases	2

24. M/J 20/P22/Q3/c

3(c)(i)	value for the activation energy for the reaction A (no enzyme) compared to reaction B (with enzyme) value / range of values that are more (+) 29 kJ mol ⁻¹	1
3(c)(ii)	value for the enthalpy change for reaction A (no enzyme) compared to reaction B (with enzyme) -14 kJ mol ⁻¹	1
3(c)(iii)	M1 show the energy of the reactants > products AND label ΔH using the predicted value given in (ii) M2 show activation energy 'hump' AND label using the value given in (i) 	2

25. MAR 20/P22/Q1/c

1(c)	high(er) activation energy / heating overcomes activation energy	1
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26. MAR 20/P22/Q2/e(i)

2(e)(i)	in the same phase / state	1
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27. O/N 19/P22/Q4/d(ii)

4(d)(ii)	heterogeneous	1
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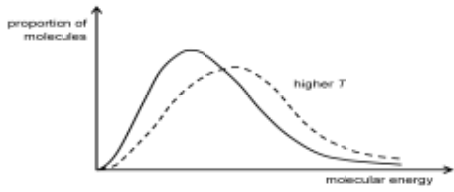
28. M/J 18/P21/Q1/a,c,d(i)

1(a)(i)	(It is a substance that) speeds up a reaction	1
	(by creating an alternative pathway / mechanism with) lower E_a	1
1(a)(ii)	(a heterogeneous catalyst is in a) different state / phase (to the reactants)	1
1(c)	1 = B	1
	2 = A	1
	3 = D	1
1(d)(i)	Increases rate AND explanation re collisions	1
	By increasing number / proportion of / more molecules / particles / species with $E \geq E_a$	1
	(So) increases frequency of successful collisions / more successful collisions per unit time / higher chance of successful collisions per unit time / higher proportion of successful collisions per unit time	1

29. M/J 18/P23/Q1/c(ii,iv)

1(c)(ii)	Decreasing reactant concentrations (at different time intervals / as reaction progresses)	1
	So rate decreases OR change in concentration $\div$ time taken decreases	1
1(c)(iv)	2:1 ratio in equation / SO ₂ used up more quickly (than O ₂) / 2 $\times$ SO ₂ react for every 1 $\times$ O ₂	1

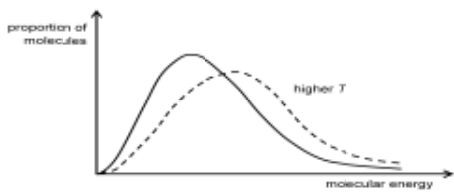
30. O/N 17/P21/Q1/b(i,ii)

1(b)(i)	general shape of the curve and peak are displaced to right of original line and starts at origin	1
	the peak is lower and curve crosses once only finishing above original line	1
		
1(b)(ii)	rate increases AND explanation in terms of collisions	1
	(at higher T) area above E_a is greater OR (at higher T) more molecules with $E \geq E_a$	1
	higher frequency of successful collisions OR more successful collisions per unit time / higher chance of successful collisions per unit time / higher proportion of successful collisions per unit time	1

31. O/N 17/P22/Q2/b(i)

2(b)(i)	(higher rate / rate increases) due to higher frequency of successful collisions	1
	more molecules / particles with $E \geq E_a$	1

32. O/N 17/P23/Q1/b(i,ii)

1(b)(i)	general shape of the curve and peak are displaced to right of original line and starts at origin	1
	the peak is lower and curve crosses once only finishing above original line	1
		
1(b)(ii)	rate increases AND explanation in terms of collisions	1
	(at higher T) area above E_a is greater OR (at higher T) more molecules with $E \geq E_a$	1
	higher frequency of successful collisions OR more successful collisions per unit time / higher chance of successful collisions per unit time / higher proportion of successful collisions per unit time	1

33. M/J 17/P22/Q3/b(i,ii)

3(b)(i)	general shape of the curve and peak are displaced to right of original and starts at origin	1
	the peak is lower and curve crosses once only finishing above original	1
3(b)(ii)	rate increases AND correct explanation in terms of 'more collisions'	1
	at higher T area above E_a is greater / more molecules with $E \geq E_a$	1
	higher frequency of successful collisions OR more successful collisions per unit time / higher chance of successful collisions per unit time / higher proportion of successful collisions per unit time	1

34. M/J 17/P23/Q3/b(i,ii)

3(b)(i)	Mark to right of original E_a	1
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3(b)(ii)	2 marks for any two points: - Benefit of using a catalyst in terms of increasing rate or economic benefit i.e. (less heat required) - Creates alternative pathway with lower E_a - More molecules with $E > E_a$	2
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35. MAR 17/P22/Q2/c(ii)

2(c)(ii)	heterogeneous (catalyst)	1
	provides an alternative reaction pathway of lower activation energy	1

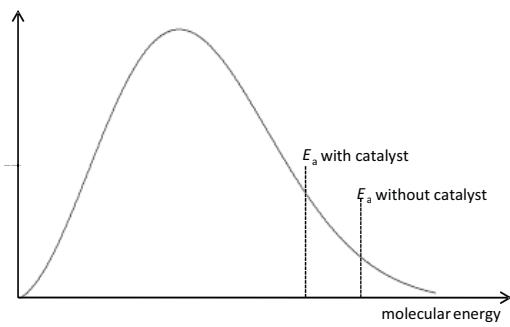
36. M/J 16/P21/Q3/b(ii)

(ii)	activation energy is high so few/no particles with $E \geq E_a$	[1] [1]	[2]
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37. M/J 16/P23/Q2/b(i,ii,iii)

(b) (i)	line from origin AND below left-hand end of original with peak to right of and lower than original crosses original once AND above right-hand end of original AND above energy axis	[1] [1]	[2]
(ii)	(curves show) more molecules with $E > E_a$ (at higher T) so greater frequency of successful (owtte) collisions / more successful (owtte) collisions per unit time	[1] [1]	[2]
(iii)	catalysed E_a shown to left of original on horizontal axis so more molecules with $E > E_a$ (in presence of catalyst)	[1] [1]	[2]

38. MAR 16/P22/Q3/c

(c)	 M1 = correct Boltzmann curve M2, M3 any 2 from: - line for both E_a values or statement in text that catalyst lowers E_a (catalyst) increases proportion/number of molecules/particles with energy $\geq$ activation energy - so more frequent successful collisions	[1] [1] [1]	[3]
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